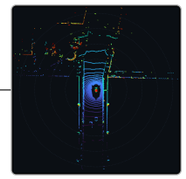
DSB perception: BEVFusion (detection + BEV segmentation) and FlashOCC (occupancy)

BEVFusion: camera Swin-T + LSS/DepthLSS, LiDAR SparseEncoder, ConvFuser; det head TransFusion, seg head 6-class BEV. FlashOCC: ResNet-50 + LSS view transform + 2D BEV encoder + channel-to-height occupancy head.



LiDAR point cloud (BEV)

BEVFusion | camera + LiDAR

Swin-T
+ GeneralizedLSSFPN

DepthLSS / LSS
view transform

camera BEV
80×180×180

Extrinsics / aug
camera2lidar · lidar2ego
img_aug · lidar_aug

geometry

Voxelize
0.075 m (xy) · 0.2 m (z)

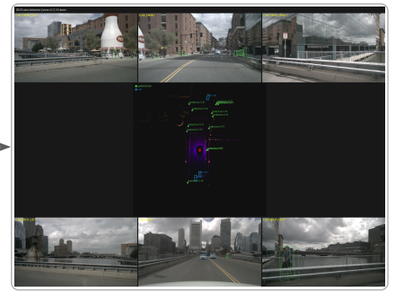
SparseEncoder
spconv

LiDAR BEV
256×180×180

ConvFuser
80+256 → 256

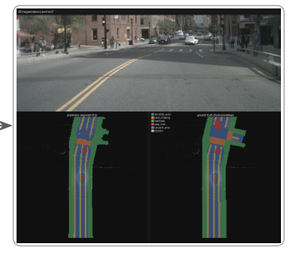
SECOND + SECONDFPN
512×180×180

TransFusion head
3D detection



detection: 6 cameras + centered BEV

BEV seg head
6-class map



BEV segmentation: pred vs GT



6 cameras (shared) · 7-tuple geometry
for FlashOCC

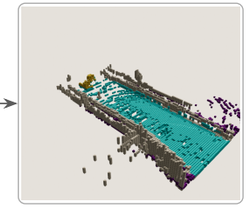
ResNet-50
+ CustomFPN

LSSViewTransformer
64×200×200

CustomResNet
+ FPN_LSS
256×200×200

BEVOCCHead2D
18 × 16 channels

FlashOCC | camera only



occupancy: 200×200×16

Grids: det 0.075×0.075×0.2 m voxel / 180×180 BEV; seg 0.1×0.1×0.2 m voxel / 128×128 BEV (+ 200×200 map @ 0.5 m); occ 0.4 m / 200×200×16 occupancy. BEVFusion uses LiDAR + 6 cameras; FlashOCC is camera-only.